Autonomous Cognitive Engine for Deep Research and Long-Horizon Tasks

Multi-Agent Research System using LangGraph & LangSmith

Presented by

Annam Lasya

PROJECT OVERVIEW

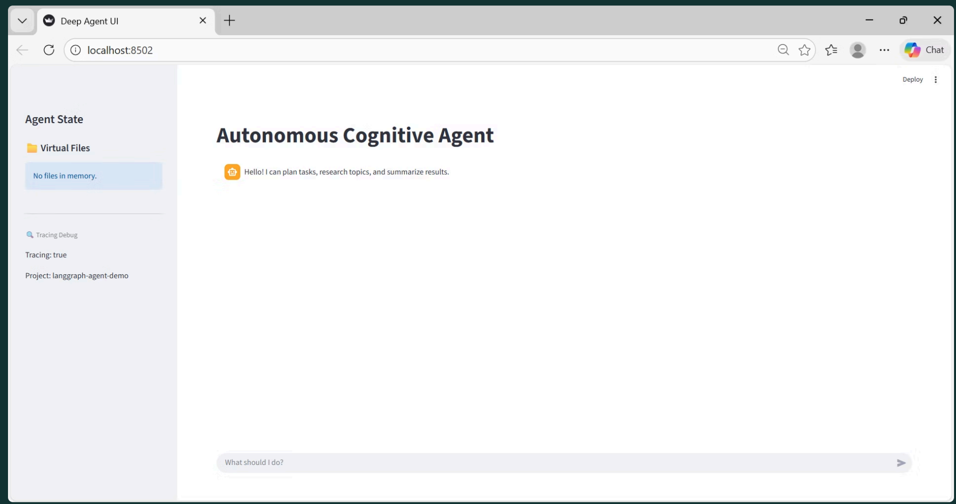
Multi-Agent Research System

This project implements a multi-agent research system using:

LangGraph (state-driven execution)

LangChain (LLM orchestration)

LangSmith (tracing) Streamlit (UI)



Goal: Clear separation of planning, research, search, and summarization, while keeping internal reasoning hidden from UI and fully traceable.

CHALLENGE

The Problem with Traditional Agents

- Single-agent design mixes planning, reasoning, and execution
- Lack of observability → reasoning and tool usage are hidden
- Difficult to debug or audit workflows
- Poor handling of long-horizon tasks and complex dependencies
- No structured memory for intermediate results

Key Takeaway: Traditional agents do not separate responsibilities, making them **opaque, brittle, and hard to scale**.

SOLUTION

A Modular Multi-Agent Approach

Multi-Agent Design

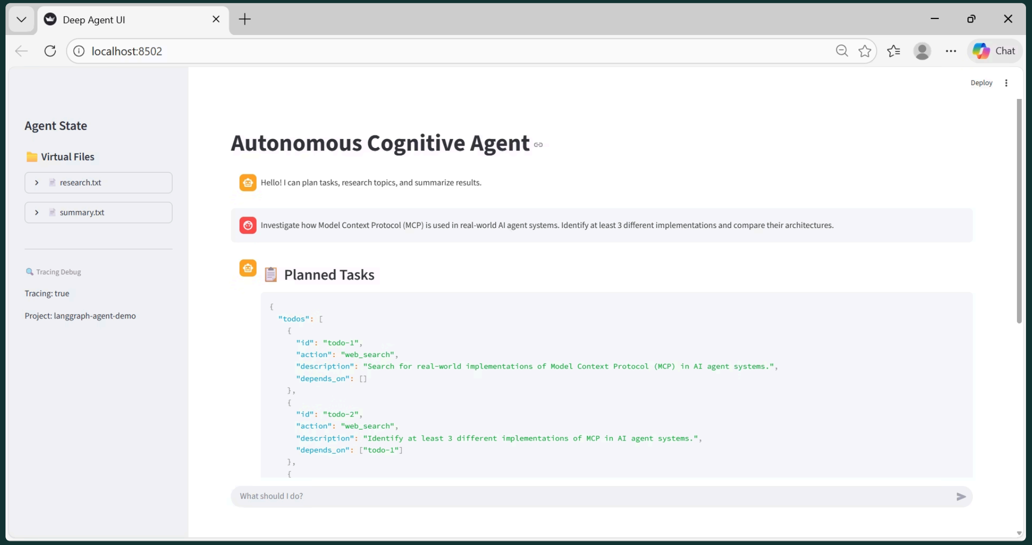
Specialized agents (Planner, Research, Search, Summarizer) work together in a deterministic, state-driven system.

Role Separation

Each agent has a clear, independent responsibility.

Tool-Based Execution

All actions use traceable tools, artifacts are stored in VFS, and LangSmith logs every step.



ARCHITECTURE

System Architecture & Flow

Execution Flow

1. User query → Planner TODOs
2. Research agent orchestrates tools
3. Search agent performs web search
4. Summarizer generates final output
5. Results stored in VFS, shown in UI

Key Features & Capabilities

- **Planner Agent:** Converts user queries into structured TODOs for controlled execution
- **Search & Research Agents:** Parallel information retrieval and tool coordination
- **Summarizer Agent:** Generates coherent final summaries
- **Virtual File System (VFS):** Shared memory for intermediate and final outputs
- **LangSmith Tracing:** Full observability of agent decisions, tool calls, and state transitions
- **Streamlit UI:** Clean interface showing only results and sidebar files

Technology Stack & Tools

- **Tavily Search API**
  Provides enterprise-grade web search with semantic ranking, enabling accurate and relevant research results.

- **OpenRouter LLM**
  Offers a single gateway to multiple large language models, allowing flexible model selection and execution.

- **Traceable Tools**
  All agent actions and tool calls are fully logged, ensuring transparency, observability, and easy debugging.

VFS ARCHITECTURE

Virtual File System Design

Shared Agent Memory

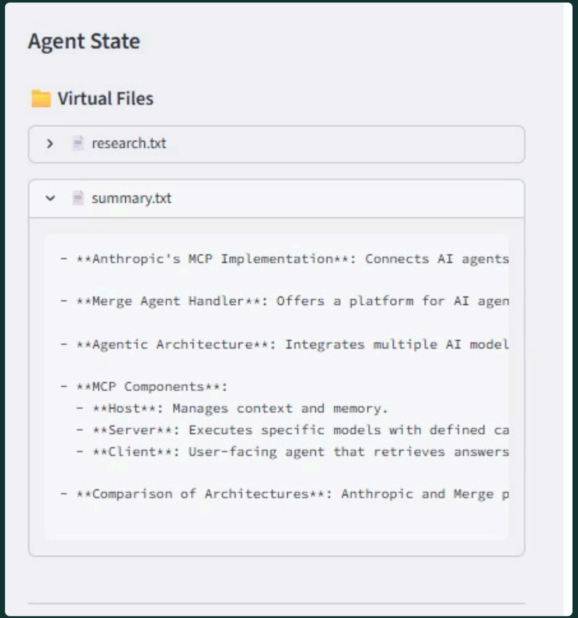
Centralised state repository enabling seamless data exchange between autonomous agents

todos.json Structure

Structured task management with status tracking, dependencies, and priority queuing

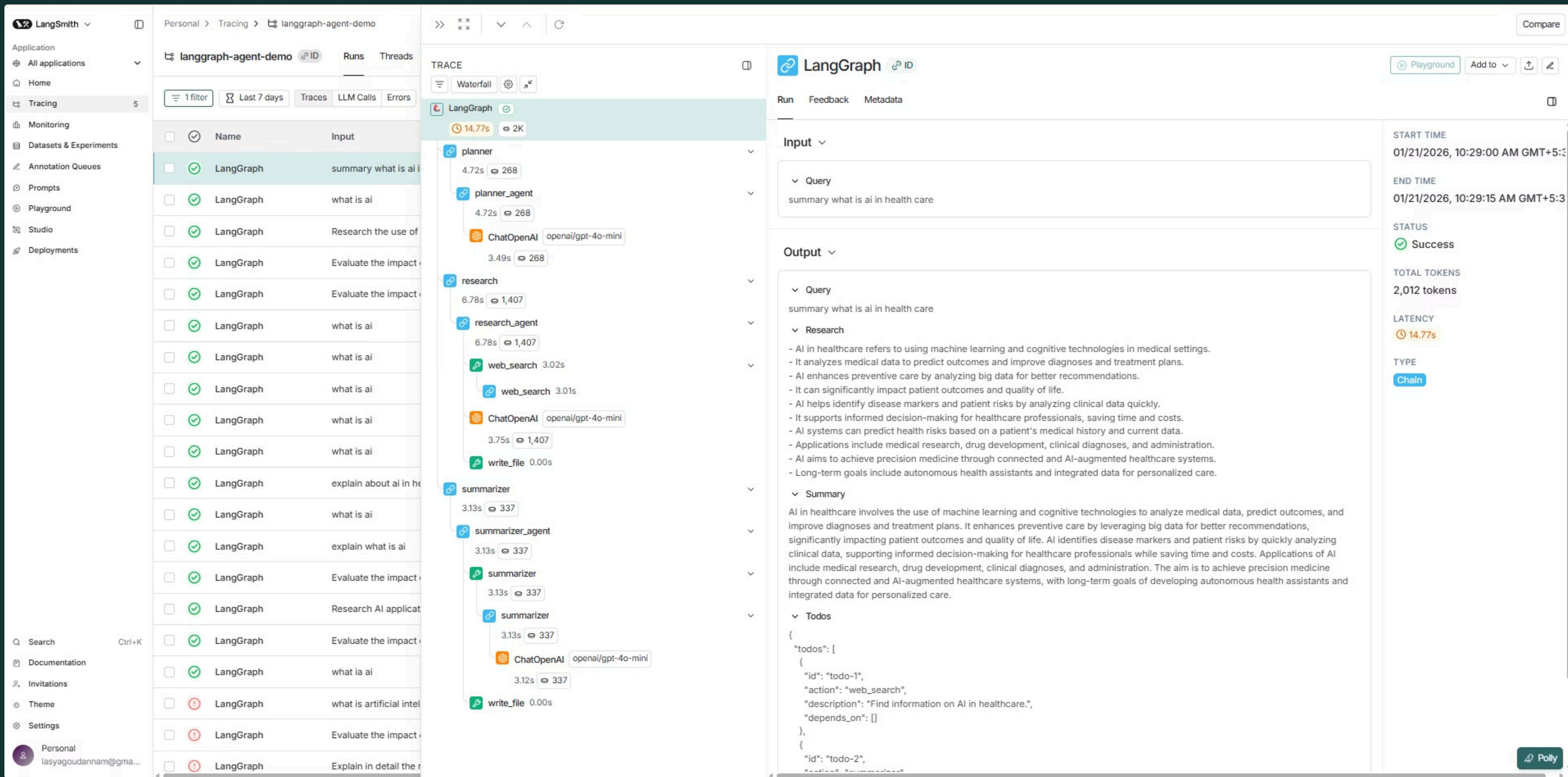
summary.md Output

Markdown-formatted final deliverable containing synthesised research findings and insights



OBSERVABILITY

LangSmith Tracing & Monitoring



LangSmith provides production-grade tracing, enabling rapid debugging, performance optimisation, and continuous improvement of agent behaviours.

Challenges And Future Scope

Challenges

- Coordinating multiple agents smoothly
- Maintaining consistent shared state
- Mapping tasks to correct tools
- Debugging & tracing agent actions
- Handling concurrent requests

Future Scope

- Fully autonomous task execution
- Integrate more AI models
- Personalized user workflows
- Automatic data visualizations
- Multi-language & cloud support

THANK YOU...

Q/A